Week 6

Introduction

In the rapidly evolving landscape of electric vehicle charging infrastructure and grid integration, the efficient and reliable conversion of electrical power is not just a convenience but a fundamental necessity. At the heart of these systems lie sophisticated power electronics, particularly three-phase AC-DC converters and isolated DC-DC converters. While the basic operation and component sizing of these converters are essential, achieving optimal performance, grid compatibility, and user safety demands advanced closed-loop control strategies. This week's lessons delve into the intricate world of controlling these critical power conversion stages.

The challenge with three-phase systems stems from the inherent interdependencies of their currents, which makes direct, independent control difficult. We explore powerful mathematical transformations that decouple these quantities, enabling precise regulation of output voltage and ensuring unity power factor operation—vital for grid stability and energy efficiency. Furthermore, we revisit isolated DC-DC converters, understanding their crucial role in providing galvanic isolation for safety and precise battery charging, particularly in electric vehicle supply equipment (EVSE). By mastering these techniques, we can design systems that not only deliver stable power but also interact harmoniously and safely with the electrical grid.

Core Concepts

Term	Definition	Significance
`Unity Power Factor (UPF)`	A condition where the current drawn from the AC source is perfectly in phase with the voltage, resulting in minimal reactive power exchange.	Essential for efficient power transfer, reducing grid losses, and minimizing harmonic distortion, ensuring the converter acts as a purely resistive load.
`Space Vector`	A single rotating vector that represents the combined instantaneous information of three-phase quantities (voltages or currents) in a two-dimensional plane.	Simplifies the analysis and control of three-phase systems by consolidating three interdependent quantities into one rotating vector, making it easier to visualize and manipulate.

Term	Definition	Significance
`DQ Reference Frame`	A two-axis coordinate system (D and Q axes) that rotates synchronously with the space vector of the three-phase quantities, often aligned with the supply voltage.	Transforms AC quantities into DC quantities under steady-state, simplifying control and enabling independent regulation of active and reactive power components.
`PI Controller`	A Proportional-Integral controller, a common feedback control loop mechanism that calculates an error value as the difference between a measured process variable and a desired setpoint.	Highly effective for controlling DC quantities in the DQ frame due to its ability to provide infinite gain at DC, ensuring zero steady-state error.
`Feedforward Control`	A control strategy that anticipates and compensates for disturbances or known system dynamics before they affect the output.	Used to cancel out cross-coupling terms and known voltage components in the DQ model, decoupling the D and Q axes for independent control.
`Phase-Locked Loop (PLL)`	An electronic circuit that generates an output signal whose phase is related to the phase of an input signal, used to synchronize with the grid voltage.	Crucial for aligning the DQ reference frame with the grid voltage vector, enabling accurate control of active and reactive power.
`Isolated DC-DC Converter`	An electronic circuit that converts a DC voltage from one level to another while providing electrical isolation between the input and output.	Essential for safety (galvanic isolation) and precise control of battery charging parameters in EVSEs.
`Galvanic Isolation`	A principle of isolating functional sections of electrical systems to prevent	Provides critical safety by protecting users from potential electrical hazards

Term	Definition	Significance
	current flow between them, often achieved using transformers.	from the grid during EV charging.
`High-Frequency Transformer`	A transformer designed to operate efficiently at switching frequencies much higher than the utility line frequency (50/60 Hz).	Significantly reduces the size and weight of the transformer, enabling compact and power-dense isolated DC-DC converters.
`Core Saturation`	A condition where a magnetic core can no longer support an increase in magnetic flux density, leading to a sharp drop in inductance.	Must be avoided in transformers and inductors as it causes high currents, increased losses, and potential component damage.

Detailed Analysis

1. Foundations of Three-Phase AC-DC Converter Control

The journey into closed-loop control for three-phase AC-DC converters begins with understanding the fundamental design and control objectives, as discussed in **Closed loop control of three-phase AC-DC converter-I**. Before implementing control, critical components like inductance, capacitance, and switch ratings must be appropriately sized. The primary control objectives are to maintain a stable DC output voltage (V0) and achieve unity power factor (UPF) operation, meaning the input current is in phase with the supply voltage.

A significant challenge in three-phase systems is the inherent dependency of phase currents; their instantaneous sum is always zero, preventing independent control of each phase. To overcome this, the concept of a `Space Vector` is introduced, representing the combined effect of three-phase quantities as a single rotating vector in a two-dimensional plane. This space vector can then be transformed into different reference frames:
- **Stationary Reference Frame (Alpha-Beta)**: This frame uses two fixed, orthogonal axes (alpha and beta) to represent the three-phase quantities as two independent AC components.
- **Rotating Reference Frame (DQ)**: This frame rotates synchronously with the space vector. By aligning the direct (D) axis with the supply voltage space vector, the AC quantities are transformed into constant DC quantities (VD, VQ) under

steady-state conditions. This transformation is crucial because it allows for the use of conventional PI controllers, which are highly effective for regulating DC values, to independently control active and reactive power components.

2. Deriving the DQ-Axis Model and Decoupling Challenges

Building on the foundational concepts, **Closed loop control of three-phase AC-DC converter-II** delves into the mathematical derivation of the DQ-axis model. The control objectives remain output voltage regulation and unity power factor. The transformation process involves converting three-phase quantities to the alpha-beta stationary frame and then to the DQ rotating frame. A key strategy is to align the d-axis with the supply voltage space vector (Vs), which simplifies the model by making the q-component of the supply voltage (Vsq) zero.

The system's dynamics are initially described by a KVL equation in the stationary alpha-beta frame. This equation is then transformed into the DQ frame by multiplying each space vector by `e^(-jtheta)`, where `theta` is the angle of the rotating frame. A critical step involves differentiating the product of the current space vector and the rotational term, which introduces additional components. This derivation yields the following coupled DQ-axis models:
- **D-axis Model**: `L * d/dt (Isd) + Rs * Isd = Vsd - V_converter_D + omega * L * Isq`
- **Q-axis Model**: `L * d/dt (Isq) + Rs * Isq = Vsq - V_converter_Q - omega * L * Isd`
These equations reveal **`Cross-Coupling Terms`** (e.g., `omega * L * Isq` and `omega * L * Isd`) that link the d-axis and q-axis dynamics. These terms indicate that a change in one axis affects the other, posing a challenge to achieving true independent control. Therefore, these cross-coupling terms must be actively compensated for to enable precise regulation of active (related to Isd) and reactive (related to Isq) power.

3. Advanced Control Strategies for Three-Phase AC-DC Converters

Closed loop control of three-phase AC-DC converter-III elaborates on the advanced control strategies required to manage the DQ-axis model effectively. The primary objectives of output voltage regulation and unity power factor are achieved by transforming three-phase quantities into the synchronous rotating DQ frame. For UPF, the q-axis current (ISQ) is set to zero, ensuring the current space vector aligns with the voltage space vector (which is aligned with the D-axis). The magnitude of the current drawn from the grid is then controlled by adjusting the D-axis current (ISD).

The converter itself introduces dynamics and delays. A three-phase half-bridge converter, typically using Sinusoidal Pulse Width Modulation (SPWM), has an inherent `Average Time Delay (TD)` approximated as half of the switching period (TSW/2). This delay must be accounted for in controller design. To address the cross-coupling terms and supply voltage components in the DQ model, `Feedforward Control` is employed. Feedforward terms are added to the control signals (Vconverter D, Vconverter Q) to cancel these undesirable effects, effectively decoupling the D and Q axes.

The overall control system typically uses a hierarchical structure:
- **Outer Voltage Control Loop**: Compares the desired DC output voltage with the measured voltage. A PI controller generates the reference for the D-axis current (ISD_reference), as ISD controls active power and thus output voltage.
- **Inner Current Control Loops**: Separate PI controllers regulate ISD and ISQ. For UPF, ISQ_reference is set to zero. These controllers, combined with the feedforward terms, generate the converter voltage commands (Vconverter D, Vconverter Q). Since the DQ quantities are DC, PI controllers are highly effective, providing infinite gain at DC for zero steady-state error.

4. Revisiting Isolated DC-DC Converters

Shifting focus from grid-side control, **Revisiting Isolated DC-DC converters-I** introduces the crucial role of isolated DC-DC converters, particularly in EV charging. The power conversion unit in an EVSE typically consists of an AC-DC stage (for PFC and voltage regulation) followed by an isolated DC-DC converter. This second stage is vital for:
- **Output Voltage and Current Control**: Precisely regulating power to the EV battery.
- `Galvanic Isolation`: Providing electrical separation between the grid and the battery for critical user safety.

The need for `High-Frequency Transformer` isolation is emphasized. According to Faraday's Law, the core area of a transformer is inversely proportional to its operating frequency ($A \propto 1/f$). Operating at high frequencies (tens to hundreds of kHz) allows for significantly smaller, lighter, and more compact transformers compared to bulky 50/60 Hz transformers, which is essential for power density in EVSEs.

The `Forward Converter` is presented as a basic isolated topology. Its operation involves:
- **DTS Period (Switch ON)**: Input voltage is applied to the primary, energy is transferred to the secondary, and the output inductor current rises. The transformer flux increases.

- **(1-D)TS Period (Switch OFF)**: The main switch turns off, and a reset winding ensures the transformer core's flux returns to zero to prevent `Core Saturation`. The output inductor freewheels.
- **Volt-Second Balance**: For the output inductor, this principle yields the output voltage relationship: Vo = n * Vin * D.
However, the forward converter has several disadvantages, including the risk of core saturation, limited switching frequency, lossy reset mechanisms, limited core utilization (only one quadrant of the B-H curve), and high voltage stress on the main switch, which restrict its use to lower power applications.

5. Designing Robust Control Loops

Finally, **Closed loop control of three-phase AC-DC converter-IV** brings together the control concepts by detailing the design of robust control loops. The objectives remain output voltage regulation and unity power factor, achieved through the DQ reference frame with Vsq=0 and ISQ=0. The DQ axis model, with its cross-coupling terms, necessitates feed-forward compensation to achieve decoupled control.

The design of the inner current loops (for ISD and ISQ) utilizes PI controllers. A common approach is pole-zero cancellation, where the zero of the PI controller is set to cancel the pole of the plant (L-Rs series impedance). This simplifies the dynamics, allowing for the calculation of proportional gain (KPC) and integral gain (KIC) based on desired damping (e.g., ζ=0.707) and system parameters (L, Rs, converter gain G, sensor gain K1, and time constants TD, T1).

The outer voltage loop is designed to be significantly slower than the current loops. It also employs a PI controller (KPV, KIV). Bode plot analysis is used to ensure stability and a good phase margin (e.g., 37 degrees) by strategically placing the controller's zero and considering system poles. The 'K' factor, derived from power balance, relates the converter's output current to the d-axis current. Sensor gains (K1 for current, K2 for voltage) are determined by scaling physical measurements to microcontroller-compatible voltage levels.

Crucially, the entire system relies on accurate grid synchronization, which is achieved by a `Phase-Locked Loop (PLL)`. The PLL senses grid voltages, transforms them to the DQ frame, and uses a PI controller to drive the VSQ component to zero. The output of this controller, combined with a feed-forward term for nominal grid frequency, provides the angular speed (ω). Integrating ω yields the phase angle (θ), which is essential for all ABC-to-DQ and DQ-to-ABC transformations, ensuring the converter's operation is perfectly synchronized with the grid. This comprehensive approach is known as vector control.

Key Takeaways

- [ ] Three-phase AC-DC converters require closed-loop control to regulate output voltage and achieve unity power factor.
- [ ] Space vector and DQ frame transformations are essential to decouple three-phase quantities for independent control.
- [ ] Cross-coupling terms in the DQ model necessitate feedforward compensation for effective decoupling of d-axis and q-axis control.
- [ ] Hierarchical control structures, comprising an outer voltage loop and inner current loops, are used for precise regulation.
- [ ] PI controllers are highly effective for controlling the DC quantities in the DQ frame, ensuring zero steady-state error.
- [ ] A Phase-Locked Loop (PLL) is crucial for synchronizing the converter's control frame with the grid voltage, enabling accurate transformations.
- [ ] Isolated DC-DC converters provide critical galvanic isolation for safety and precise output control in applications like EV charging.
- [ ] High-frequency transformers are fundamental to isolated DC-DC converters, enabling compact and power-dense designs.
- [ ] Basic isolated topologies like the forward converter have limitations such as core saturation risk and high switch stress, restricting their use to lower power levels.
- [ ] Robust controller design involves pole-zero cancellation, Bode plot analysis, and careful parameter tuning for stability and performance.

Conclusion

This week's exploration into the closed-loop control of three-phase AC-DC converters and the fundamentals of isolated DC-DC converters highlights the sophisticated engineering required to meet the demands of modern power systems. From the intricate mathematical transformations that decouple interdependent three-phase quantities to the meticulous design of PI controllers and feedforward compensation, every aspect contributes to achieving precise output voltage regulation and unity power factor operation. The indispensable role of the Phase-Locked Loop in synchronizing with the grid underscores the need for seamless integration into the electrical network.

Furthermore, understanding isolated DC-DC converters, particularly their ability to provide galvanic isolation through high-frequency transformers, is paramount for safety and efficient power delivery in applications like electric vehicle charging. While simpler topologies like the forward converter offer foundational insights, their

limitations drive the development of more advanced, high-power solutions. These advanced control methodologies and converter designs are not merely academic exercises; they are the backbone of efficient, reliable, and safe power electronics that are fundamental to the advancement of sustainable energy infrastructure, from renewable energy integration to the widespread adoption of electric vehicles.